

1. Find an inverse of a modulo m for each of these pairs of relatively prime integers.
 1. $a = 4, m = 9$
 2. $a = 19, m = 141$
 3. $a = 55, m = 89$
 4. $a = 89, m = 232$
2. Show that if m is an integer greater than 1 and $ac \equiv bc \pmod{m}$, then $a \equiv b \pmod{\frac{m}{\gcd(c, m)}}$.
3. Use Fermat's little theorem to find $7^{121} \bmod 13$.
4. Use Fermat's little theorem to show that if p is prime and $p \nmid a$, then a^{p-2} is an inverse of a modulo p .
5. Find the greatest common divisor (gcd) and least common multiple (LCM) of the following pairs of numbers using their prime factorizations:
 1. 72 and 120
 2. 168 and 180
6. Find some composite number n such that there exists an integer a such that Fermat's Little Theorem does not hold for a in the sense that, there exists an a such that $\gcd(a, n) = 1$ and

$$a^{n-1} \not\equiv 1 \pmod{n}.$$